

Multi-Threshold Pass-Count Preservation under Bounded Score Perturbations

Research Architecture Runtime
operator/multi-threshold

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Abstract

Scalar threshold $A_T(x) = \mathbf{1}\{x \geq T\}$ is invariant outside the 2ε unstable band when $|x' - x| \leq \varepsilon$. Lean `LEAN_FULLL`.

This theorem is: Primitive.

1 Problem

The operator returns the binary indicator $A_T(x) = 1$ iff $x \geq T$ (equality passes).

2 Stability notion

Perturbation: $|x' - x| \leq \varepsilon$. Pass buffer $x \geq T + \varepsilon \Rightarrow A_T(x') = 1$; fail buffer $x < T - \varepsilon \Rightarrow A_T(x') = 0$.

3 Definitions

Definition 1. $A_T(x) \Leftrightarrow x \geq T$.

Assumptions. $\varepsilon \geq 0$ and $|x' - x| \leq \varepsilon$.

4 Theorem

Theorem 2. *If $|x' - x| \leq \varepsilon$, then $x \geq T + \varepsilon \Rightarrow x' \geq T$ and $x < T - \varepsilon \Rightarrow x' < T$.*

5 Intuition

Moving x by at most ε cannot cross T from outside the ε -neighborhood.

6 Examples

Example 3. $T = 3$, $x = 5$, $\varepsilon = 0.2$: $x \geq T + \varepsilon$, so every x' in the ball still passes.

7 Proof

From $|x' - x| \leq \varepsilon$ one has $x' - \varepsilon \leq x \leq x' + \varepsilon$. If $x \geq T + \varepsilon$ then $x' \geq x - \varepsilon \geq T$. If $x < T - \varepsilon$ then $x' \leq x + \varepsilon < T$ (Theorem 2).

8 Formal statement

Lean module: `Research.Operators.MultiThreshold.Preservation`. Source file: `lean/Research/Operators/MultiThreshold.Preservation`.
Theorems: `multi_threshold_preservation`, `multi_threshold_sharpness`. Certificate: `lean/certificates/multi_threshold_preservation.cert`.
Status: `LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).

9 Proof dependencies

- Absolute-value ball arithmetic only.
- No other operator theorems.

10 Consequences

Primitive scalar gate used by multi-threshold, abs-threshold, interval membership, and noisy threshold variants.